TEXORA · COURSE NOTES · LECTURE SHEET

Colloids and Colloidal Solution

The middle kingdom of dispersions — particle sizes between solution and suspension, the Tyndall effect, Brownian motion, charge and coagulation, and why dyes and finishes are colloids.

CHE-1105 · CHEMISTRY-I · TOPIC 8 · LEVEL 1 TERM 1

1. A matter of size

When one substance is dispersed in another, everything depends on the particle size of the dispersed phase:

SYSTEM	PARTICLE SIZE	BEHAVIOUR
True solution	below ~1 nm (ions, small molecules)	Homogeneous; passes membranes and filters; never settles; no Tyndall effect.
Colloidal solution (sol)	~1 to ~1000 nm	Appears homogeneous but is not; passes filter paper, not membranes; scatters light (Tyndall); settles extremely slowly or never.
Suspension	above ~1000 nm	Visible particles; retained by filters; settles on standing.

A colloidal system has two parts: the **dispersed phase** and the **dispersion medium**. Named combinations fill daily life — liquid in gas is an aerosol (fog, spray), gas in liquid a foam, liquid in liquid an emulsion (milk, finishing oil emulsions), solid in liquid a sol (gold sol, starch, most dye dispersions), solid in solid a solid sol (coloured glass). Milk, blood, paint, fog and the dyebath are all colloids or close relations.

2. Two families: lyophilic and lyophobic

FEATURE	LYOPHILIC (SOLVENT-LOVING)	LYOPHOBIC (SOLVENT-HATING)
Nature	Giant molecules themselves (starch, gelatin, proteins, cellulose derivatives)	Aggregates of many small molecules/atoms (gold, sulphur, metal hydroxides)

Preparation	Form spontaneously on contact with solvent	Need special methods (grinding, peptisation, controlled reaction)
Stability	Very stable; heavily solvated	Kept stable only by particle charge; easily coagulated
Reversibility	Reversible — dried residue re-forms the sol on adding solvent	Irreversible — once coagulated, does not re-disperse by solvent alone
Effect of electrolytes	Tolerate moderate amounts; salted out only by large quantities	Coagulated by traces of electrolyte

3. The signature properties

- **Tyndall effect:** a beam of light passing through a sol is scattered by the particles and becomes visible as a bright cone — the test that distinguishes a colloid from a true solution (whose beam stays invisible), and the reason headlight beams show in fog.
- **Brownian motion:** under the ultramicroscope the particles dance in ceaseless zig-zag, bombarded by the medium's molecules — the first direct visible evidence of molecular motion, and the reason sols do not settle: the bombardment keeps re-suspending them against gravity.
- **Electrical charge and electrophoresis:** colloidal particles carry charge (from adsorbed ions), all particles of a sol the same sign — the mutual repulsion is the lyophobic sol's main defence against joining. In an electric field the particles migrate to the opposite electrode (**electrophoresis**), a property used analytically and industrially (electrophoretic painting).
- **Dialysis:** membranes pass true-solution particles but hold colloids back, so a sol is purified of dissolved impurities by dialysis — the principle of the artificial kidney.

4. Coagulation — and protection against it

Add electrolyte to a lyophobic sol and the particles, their charge neutralised by the oppositely charged ions, collide and stick — the sol **coagulates** (precipitates). The **Schulze–Hardy rule** states the pattern: the coagulating power of an ion rises dramatically with its valency — for a negative sol, $Al^{3+} \gg Mg^{2+} \gg Na^{+}$ (roughly in the ratio of hundreds : tens : 1). This is why a pinch of alum clears muddy water (and why alum treats dyehouse effluent): trivalent aluminium ions collapse the negative clay and dye colloids. The reverse art is **protection**: adding a lyophilic colloid (gelatin, gum) coats lyophobic particles and shields them from electrolytes. Protective power is measured by the **gold number** — the milligrams of protective colloid that just fail to prevent a standard gold sol from coagulating with salt; the smaller the gold number, the stronger the protection (gelatin's is near 0.005–0.01, the best common protector).

5. Preparation in outline

Lyophobic sols are made by forcing matter into the colloidal size window from either side: **dispersion methods** break coarse material down (colloid mills, peptisation — shaking a fresh precipitate with a little electrolyte to charge and disperse it), and **condensation methods** build up from molecules (controlled reduction gives gold sol; hydrolysis of $FeCl_3$ gives ferric hydroxide sol; solvent exchange precipitates sulphur sol). Either way the sol must be purified by dialysis before use.

6. The textile thread

Wet processing is applied colloid science. **Disperse dyes** are quite literally colloidal dispersions of insoluble dye, milled with dispersing agents to the sub-micron range so they can enter polyester; pigment printing pastes, latex binders, silicone and softener emulsions, and starch sizes are all engineered colloids whose stability in the bath — against the electrolyte and heat of processing — is decided by the charge and protection ideas of this topic. Even effluent treatment closes the loop: colour is removed from dyehouse wastewater by coagulating its colloidal dye and solids, Schulze–Hardy at industrial scale.

7. Glossary — key terms of this topic

TERM	MEANING
Dispersed phase / medium	The particles scattered through a colloid / the substance they are scattered in.
Tyndall effect	Scattering of a light beam by colloidal particles, making the beam visible.
Brownian motion	The ceaseless zig-zag motion of colloidal particles under molecular bombardment.
Electrophoresis	Migration of charged colloidal particles in an electric field.
Coagulation	The clumping and precipitation of a sol when its particle charge is neutralised.
Schulze–Hardy rule	Coagulating power of an ion increases sharply with its valency.
Gold number	A measure of a colloid’s protective power; smaller means stronger protection.
Peptisation	Dispersion of a fresh precipitate into a sol by adding a small amount of electrolyte.

8. Key points and summary

- Colloids are dispersions of 1–1000 nm particles — between true solutions and suspensions — classified by the phases of dispersed material and medium.

- Lyophilic sols are stable, solvated and reversible; lyophobic sols survive by particle charge alone and fall to traces of electrolyte.
- Tyndall scattering, Brownian motion, electrophoresis and dialysis are the working tests and tools; coagulation follows Schulze–Hardy, and protection is measured by the gold number.
- Dye dispersions, printing pastes, finishing emulsions and effluent coagulation make this topic daily textile practice.

9. Practice

Answer in your own words; brief model answers follow.

1. Classify by dispersed phase and medium: fog, milk, whipped cream, gold sol, ruby glass.
2. A liquid shows a bright Tyndall cone but passes through filter paper unchanged. True solution, colloid or suspension — and which single observation rules out each wrong answer?
3. Why does Brownian motion prevent a gold sol from settling, while a sand suspension settles in minutes?
4. A negatively charged sol is to be coagulated. Rank NaCl, $MgCl_2$ and $AlCl_3$ by the amount needed (least first) and name the rule.
5. Gelatin has a gold number near 0.01 and starch near 25. Which is the better protective colloid, and what does the number literally measure?
6. State two differences between lyophilic and lyophobic sols concerning electrolytes, and the molecular reason lyophilic sols are stable.
7. A dialysis bag containing a starch sol with dissolved salt is placed in pure water. After a day, what is inside the bag and what has left?
8. Disperse dyes for polyester are milled with dispersing agents to sub-micron size. Explain, in colloid terms, what the dispersing agent is doing and what would happen in the hot dyebath without it.

Answer key

1. Fog — liquid in gas (aerosol). Milk — liquid in liquid (emulsion). Whipped cream — gas in liquid (foam). Gold sol — solid in liquid (sol). Ruby glass — solid in solid (solid sol).
2. A colloid. The Tyndall cone rules out a true solution (no scattering); passing the filter rules out a suspension (whose particles are retained).
3. Colloidal particles are small enough that molecular bombardment continually kicks them back up, balancing gravity; sand grains are far too massive for the bombardment to matter and sediment directly.

4. $AlCl_3$ (least) < $MgCl_2$ < NaCl (most) — the Schulze–Hardy rule: coagulating power rises steeply with the valency of the ion opposite in charge to the sol ($Al^{3+} \gg Mg^{2+} \gg Na^{+}$).
5. Gelatin — the smaller gold number means stronger protection. It is the milligrams of protector that just fail to stop 10 mL of standard gold sol turning red-to-blue on adding 1 mL of 10% NaCl.
6. Lyophilic sols tolerate moderate electrolyte (salting out only at high concentration) while lyophobic sols coagulate with traces; lyophilic particles are individually solvated giant molecules, stable by solvation, not by charge alone. (Also: reversibility.)
7. The starch sol remains (colloid particles cannot cross the membrane); the dissolved salt has dialysed out into the surrounding water.
8. The agent adsorbs on the dye particles, giving them like charges and a protective layer so they repel and stay dispersed (a stabilised lyophobic system). Without it, heat and electrolytes would coagulate the dye into specks — unlevel, spotty dyeing and blocked machinery.
